



SignalsFrame

≡ Can show publicly	Yes
≡ Category	Media & Creative Technology
🌐 Country	Morocco Qatar Saudi Arabia Spain UAE USA
🔗 Industry	AI media, video editing, creator tools, browser-based production software
≡ Main features	- - -
🔗 Project link	https://www.signalsframe.com/
≡ Project type	Client Work / Internal Product / Productized Prototype
≡ Proof available	Public link: Screenshots: Demo: Testimonials: Metrics:
≡ Status	In Development
≡ Technical stack	- - -

I'll audit this repository systematically. Let me start by exploring its structure and contents. Now let me get more information about the project structure and core files: # Project Repository Report

1. Project identity

- Project name: SignalsFrame
- Repository name: khairiddine-arbouch/SignalsFrame
- Project category: Media editing / Video production software
- Apparent project type: SaaS / Web application
- Public-facing or internal-facing: Public-facing (deployed to signalsframe.vercel.app)
- Main user/customer: Content creators, video editors, social media producers

2. One-sentence factual description

SignalsFrame is a browser-based video, audio, and photo editing application built with TypeScript, React, and WASM, featuring timeline-based editing, AI-powered background removal and auto-reframing, and multi-format export.

3. What the project does

- **Browser-based editing interface:** Provides a full editing environment (welcome screen, editor interface, project templates, recent projects, and sharing functionality) accessible via hash-based routing
- **Media timeline editing:** Implements timeline-based track management with clip handling, playback synchronization, and frame-by-frame control
- **Video/audio/photo processing:** Supports import and export of multiple media formats with frame rate and resolution management
- **AI-powered features:** Includes background removal with blur/color/image/transparent background modes, auto-reframing for social media platforms (TikTok, YouTube, Instagram, etc.), and emotion/sentiment analysis on video content
- **Real-time playback:** Uses WebCodecs and WebGPU for high-performance video rendering and playback
- **Cloud storage and collaboration:** Integrates Clerk for authentication, Convex for backend, and AWS S3 for media storage with presigned URLs

4. Confirmed implemented features

Feature	What it does	Evidence file paths	Confidence: High / Medium / Low
Authentication	User sign-in and session management via Clerk	<code>apps/web/src/main.tsx</code> , <code>apps/web/package.json</code> (Clerk React dependency), <code>apps/web/src/components/cloud/SignInRequiredDialog.tsx</code>	High
Project creation	Create new projects with custom dimensions (preset or freeform) for specific social media platforms	<code>apps/web/src/App.tsx</code> lines 74-113, preset dimensions map	High
Welcome/template interface	Landing screen with tabs for templates, recent projects, and project creation	<code>apps/web/src/App.tsx</code> lines 147-154, route handling	High

Feature	What it does	Evidence file paths	Confidence: High / Medium / Low
Video timeline	Core timeline abstraction with clips, tracks, playback state	<code>packages/core/src/timeline</code> directory, <code>packages/core/src/types/timeline</code>	High
FFmpeg integration	Video and audio transcoding/processing via <code>@ffmpeg/ffmpeg</code>	<code>packages/core/package.json</code> , build scripts	High
WASM modules	Beat detection, WAV encoding, and FFT (Fast Fourier Transform) processing	<code>packages/core/src/wasm/</code> directory, build:wasm scripts in <code>package.json</code>	High
Background removal	AI-powered background segmentation with multiple render modes (blur, solid color, image, transparent)	<code>packages/core/src/ai/background-removal-engine.ts</code> full implementation	High
Auto-reframing	Intelligent crop generation for multiple platforms with face detection and smooth keyframe interpolation	<code>packages/core/src/ai/auto-reframe-engine.ts</code> full implementation	High
Emotion/sentiment analysis	Timeline analysis for emotion and sentiment prediction with chunked processing	<code>packages/core/src/ai/timeline-analysis.ts</code> , Inngest job queue integration	High
Video export	Multi-codec export with quality presets, upscaling options, and capability detection	<code>packages/core/src/export/</code> directory, <code>ExportEngine</code> class	High
Social media presets	Predefined dimensions/frame rates for TikTok, Instagram (reels/feed/stories), YouTube, Facebook, Twitter/X, LinkedIn	<code>apps/web/src/App.tsx</code> lines 35-41, <code>SOCIAL_MEDIA_PRESETS</code> from core	High
Project recovery	Auto-recovery of unsaved projects from IndexedDB	<code>apps/web/src/components/welcome/RecoveryDialog.tsx</code> , <code>useProjectRecovery</code> hook	High
Share/collaboration links	Share functionality with unique shareId routing	<code>apps/web/src/App.tsx</code> lines 154-162, <code>SharePage</code> component	Medium
Analytics	PostHog integration for user event tracking	<code>apps/web/src/main.tsx</code> lines 10-19	High

5. Partially implemented or unclear features

Feature / claim	Why it is unclear or partial	Evidence file paths	What needs confirmation
Face detection	Auto-reframe uses skin color heuristics instead of	<code>packages/core/src/ai/auto-reframe-engine.ts</code> lines 309-380 (<code>detectFacesSimple</code> , <code>isSkinColor</code>)	Whether actual ML model integration is

Feature / claim	Why it is unclear or partial	Evidence file paths	What needs confirmation
	ML model; fallback detection if no faces found	functions)	planned or if heuristic is final solution
Background removal model	No external ML model referenced; uses simple luminance/saturation thresholding	<code>packages/core/src/ai/background-removal-engine.ts</code> lines 149-175	Whether production accuracy is acceptable or if integration with real segmentation model is planned
Timeline analysis inference	Type definitions and job status management present, but actual inference calls not visible in codebase	<code>packages/core/src/ai/timeline-analysis.ts</code> (types only)	Whether Modal AI or another provider handles actual emotion/sentiment inference
Real-time collaboration	No multi-user editing, locking, or conflict resolution visible	<code>apps/web/src/App.tsx</code> , <code>apps/web/src/stores/</code>	Whether multiplayer editing is in scope
Mobile support	MobileBlocker component prevents mobile access; unclear if intentional or WIP	<code>apps/web/src/components/MobileBlocker.tsx</code>	Whether mobile editing is planned or deliberately out of scope

6. Mocked, hardcoded, or demo-only parts

Area	What appears mocked/demo-only	Evidence file paths	Risk if claimed publicly
Face detection	Heuristic-based skin color detection with fallback default face region if detection fails	<code>packages/core/src/ai/auto-reframe-engine.ts</code> lines 322-341	Claiming "ML-powered face detection" would be misleading; actual capability is color-based heuristic
Background removal segmentation	Simple color threshold-based mask without deep learning model	<code>packages/core/src/ai/background-removal-engine.ts</code> lines 149-175	Claiming "AI background removal" without clarifying it's algorithmic (not ML) would overstate capability
Emotion/sentiment analysis	Only type definitions and job structure; no actual inference implementation visible	<code>packages/core/src/ai/timeline-analysis.ts</code>	Claiming this feature is "implemented" without visible inference code risks broken feature on launch
Theme switching	Dark mode toggle removed from product; theme hardcoded to light	<code>apps/web/src/App.tsx</code> lines 55-58	Should not claim dark mode support
Mobile editing	Blocked at UI level, not backend; deliberate or incomplete is unclear	<code>apps/web/src/components/MobileBlocker.tsx</code>	Do not claim mobile support without clarification

7. Technical stack

Layer	Technologies confirmed	Evidence file paths
Frontend framework	React 18.3.1, TypeScript 5.4.5, Vite 5.3.1	<code>apps/web/package.json</code>
State management	Zustand 4.5.2	<code>apps/web/package.json</code>
UI components	Radix UI (dialog, dropdown, slider, tabs, tooltip, popover, context-menu, select), Tailwind CSS, Lucide icons	<code>apps/web/package.json</code>
Animation	Framer Motion 12.23.24, GSAP 3.14.2	<code>apps/web/package.json</code>
3D graphics	Three.js 0.182.0	<code>apps/web/package.json</code>
Backend/serverless	Convex 1.38.0 (backend-as-a-service)	<code>package.json</code> root dependencies
Task queue/workflows	Inngest 4.4.0 (async job orchestration)	<code>package.json</code> root dependencies
Authentication	Clerk 6.6.2	<code>apps/web/package.json</code>
Media processing	FFmpeg.wasm (@ffmpeg/core 0.12.6, @ffmpeg/core-mt 0.12.6, @ffmpeg/ffmpeg 0.12.15)	<code>package.json</code> , <code>packages/core/package.json</code>
Media utility library	mediabunny 1.25.3	<code>package.json</code> , <code>packages/core/package.json</code>
WebAssembly	AssemblyScript 0.27.0 for WASM compilation (FFT, WAV encoding, beat detection)	<code>packages/core/package.json</code> build scripts
Storage	AWS S3 (@aws-sdk/client-s3 3.1045.0, @aws-sdk/s3-request-presigner 3.1045.0) with presigned URLs	<code>package.json</code>
Local storage	IndexedDB via idb-keyval 6.2.2	<code>packages/core/package.json</code>
Analytics	PostHog 1.335.2	<code>apps/web/package.json</code>
Testing	Vitest 1.6.0 (unit), Playwright 1.60.0 (E2E), React Testing Library 16.0.0	<code>apps/web/package.json</code>
Code quality	ESLint 9.39.2, TypeScript strict checking	<code>apps/web/package.json</code>
Deployment	Cloudflare Pages (Wrangler 3.114.17)	<code>apps/web/package.json</code> deploy scripts
Build tools	pnpm 9.0.0 (monorepo), PostCSS, Autoprefixer	<code>package.json</code>
Service worker	Custom service worker registration	<code>apps/web/src/services/service-worker</code>

8. Routes, pages, and user flows

Route/Page	Purpose	Evidence file paths
<code>#/welcome</code>	Landing page with tabs for templates, recent projects, and project creation	<code>apps/web/src/App.tsx</code> lines 147-154, WelcomeScreen component
<code>#/templates</code>	Browse project templates	<code>apps/web/src/App.tsx</code> lines 149-150
<code>#/recent</code>	View recently opened projects	<code>apps/web/src/App.tsx</code> lines 151-152
<code>#/new</code>	Create a new project, optionally with preset (TikTok, YouTube, Instagram, etc.) or custom dimensions	<code>apps/web/src/App.tsx</code> lines 74-113, preset handling
<code>#/editor</code>	Main video/audio/photo editing interface with timeline, preview, and tool panels	<code>apps/web/src/App.tsx</code> lines 166-168, EditorInterface lazy load
<code>#/share/:shareId</code>	View shared project (read-only or collaborative view)	<code>apps/web/src/App.tsx</code> lines 154-162
Search modal (Ctrl+K)	Global search for assets/clips	<code>apps/web/src/App.tsx</code> lines 134-137, SearchModal component

Route/Page	Purpose	Evidence file paths
Script view dialog	View/edit clip or project scripts/metadata	<code>apps/web/src/App.tsx</code> lines 171-174, ScriptViewDialog component
Analyze clip range dialog	Trigger emotion/sentiment analysis on selected timeline range	<code>apps/web/src/App.tsx</code> line 186, AnalyzeClipRangeDialog component
Sign-in required dialog	Prompt for authentication when needed	<code>apps/web/src/App.tsx</code> line 184
Save to cloud required dialog	Prompt to save project before accessing cloud features	<code>apps/web/src/App.tsx</code> line 185

9. Data model and backend logic

Database/schema/entities:

- Projects (with metadata, dimensions, frame rate, timeline)
- Clips (with media references, in/out points, start time, speed, effects, text overlays)
- Tracks (video, audio, graphics, text)
- Media assets (video, audio, image files stored in S3)
- Analysis jobs (emotion/sentiment for timeline ranges)
- User data (authentication via Clerk, profile, preferences)
- Evidence: `packages/core/src/types/timeline.ts`, `packages/core/src/storage/` directory

API endpoints/server actions:

- Backend abstracted through Convex (serverless backend); specific endpoints not directly visible in repo but referenced via `convex` dependency
- Inngest handles async processing: video export, AI inference, timeline analysis
- Evidence: `package.json` dependencies on `convex` and `inngest`, `apps/web/src/hooks/useKieAIPoller.ts` (polling for analysis results)

Important business logic:

- Project creation with preset social media dimensions
- Auto-reframe analysis: frame-by-frame face detection, crop calculation with smoothing and center-bias
- Background removal: mask generation and blending for multiple render modes
- Timeline analysis chunking: break video into overlapping segments for inference (6-second max, 500ms overlap)
- Export planning and execution with codec selection and quality presets
- Project recovery from IndexedDB for crash resilience

Data freshness:

- AI analysis results tracked with freshness state (fresh/partially_stale/stale) based on clip metadata changes
- Emotion/sentiment predictions cached per timeline segment
- Evidence: `packages/core/src/ai/timeline-analysis.ts` lines 284-316

10. AI-specific analysis

AI providers and models:

- **Background removal:** Custom heuristic algorithm (luminance + saturation thresholding), not ML model. No external provider detected.
- **Auto-reframe/face detection:** Custom skin-color detection heuristic with fallback, not ML model. No external provider detected.
- **Emotion/sentiment analysis:** Structured for external inference via Modal AI (inferred from timeline-analysis.ts type names like `ModalInferenceOutput`), but actual inference code not in codebase. Likely handled server-side.

AI features implemented:

1. Background removal (algorithmic, not ML):

- Modes: blur, solid color, image background, video background, transparent
- Processes frames to generate luminance-based masks
- Applies edge blur and blends original foreground with background
- Evidence: `packages/core/src/ai/background-removal-engine.ts` full implementation

2. Auto-reframing (heuristic-based face detection):

- Detects skin regions using color thresholds
- Generates reframe keyframes for target aspect ratios (16:9, 9:16, 1:1, 4:5, 4:3, 21:9, custom)
- Supports platform presets: YouTube, TikTok, Instagram, Facebook, Twitter/X, LinkedIn
- Applies smoothing and tracking speed for natural motion
- Evidence: `packages/core/src/ai/auto-reframe-engine.ts` full implementation

3. Timeline analysis (structure only, inference external):

- Chunks video into 6-second segments with 500ms overlap
- Tracks emotion predictions (anger, disgust, fear, joy, neutral, sadness, surprise)
- Tracks sentiment predictions (negative, neutral, positive)
- Manages freshness state based on clip metadata changes
- Evidence: `packages/core/src/ai/timeline-analysis.ts`

Prompts: Not found in repo. Inference appears server-side via Modal AI or similar.

Output handling:

- Background removal outputs processed frames as ImageBitmap; integrated into export pipeline
- Auto-reframe outputs keyframe data stored in clip metadata; used during playback/export
- Timeline analysis results stored as TimelineAnalysisSegment records with emotion/sentiment predictions; cached and displayed in UI (polling via `useKieAIPoller`)

Inference calls:

- Background removal: Runs client-side on each frame during export
- Auto-reframe: Runs client-side on frame sequence during analysis
- Timeline analysis: Server-side inference orchestrated via Inngest; client polls `useKieAIPoller` hook for results
- No evidence of real production inference for emotion/sentiment; likely placeholder or server implementation not exposed in repo

11. Portfolio table improvements

Field	Content
Status	Early-stage / Beta (v0.1.0, under 2 weeks old, private repo)
Industry	Media / Video production / SaaS
Category	Browser-based video editing application
Project type	Full-stack web application with AI features
What we built	A browser-based timeline video editor with FFmpeg transcoding, real-time preview via WebCodecs/WebGPU, AI-powered background removal and auto-reframing for social media, emotion/sentiment analysis, and cloud storage integration
Client/user problem	Content creators need accessible, browser-based tools to edit video, audio, and photos for multiple social media platforms without installing desktop software
Main features	Social media preset dimensions, multi-track timeline editing, FFmpeg-powered export, background removal with multiple render modes, auto-reframing with face tracking, emotion/sentiment analysis, project templates, cloud storage with AWS S3, project recovery, shared project links
Technical stack	React + TypeScript, Vite, Zustand, Tailwind + Radix UI, Framer Motion, Three.js, FFmpeg.wasm, AssemblyScript (WASM for DSP), Convex backend, Inngest workflows, Clerk auth, AWS S3, IndexedDB, PostHog analytics, Cloudflare Pages deployment
Proof available	Deployed to signalsframe.vercel.app (requires Clerk sign-in); tests present (Vitest, Playwright) but not run/verified; repo is private
Best sales angle	"The only browser-based video editor with one-click auto-reframing for every social platform and real-time emotion analysis on your content"
What should not be claimed	Do not claim "AI-powered background removal" without clarifying it uses algorithmic detection, not ML. Do not claim ML-based face detection—it's heuristic skin-color tracking. Do not claim emotion/sentiment analysis is production-ready without confirming server-side inference is live. Do not claim mobile support. Do not claim dark mode.

12. Safe public claims

- **Browser-based editing:** The application runs entirely in the browser via React and WebAssembly.
- **Multi-format export:** Supports video, audio, and image export with codec selection.
- **Social media presets:** Includes predefined dimensions for TikTok, YouTube, Instagram (reels, feed, stories), Facebook, Twitter/X, and LinkedIn.
- **Timeline editing:** Full non-linear editing with multi-track support for video, audio, graphics, and text.
- **Background removal:** Includes background replacement with blur, solid color, image, video, or transparent modes.
- **Auto-reframing:** Automatically crops video to fit target aspect ratios with smooth keyframe interpolation.
- **Cloud storage:** Integrates AWS S3 for media storage and Clerk for user authentication.
- **Project recovery:** Auto-saves and recovers unsaved projects from local IndexedDB.
- **Shared projects:** Generate shareable links for collaborative or read-only viewing.
- **Real-time preview:** Playback powered by WebCodecs and WebGPU for high performance.
- **Emotion and sentiment tracking:** Analyzes video content for emotional and sentiment trends (if server inference is confirmed).

13. Unsafe or unproven claims

- **"AI-powered background removal"** — The implementation is algorithmic (luminance/saturation thresholding), not machine learning. Accuracy will be limited vs. ML-based solutions.
- **"ML-based face detection"** — Uses skin-color heuristics, not a trained model. Fallback triggers if no faces detected, reducing reliability.
- **"Production-ready emotion/sentiment analysis"** — Type definitions and job infrastructure exist, but actual inference not visible. Server-side implementation unclear.
- **"Mobile-first editing"** — Mobile access is explicitly blocked; unclear if intentional or WIP.
- **"Dark mode support"** — Theme hardcoded to light; dark mode toggle removed from UI.
- **"Multiplayer collaborative editing"** — No multi-user synchronization, conflict resolution, or locking mechanisms visible.
- **"Professional-grade color grading"** — Keywords in package.json mention "color-grading" but no implementation found; likely placeholder.
- **"LIVE production ready"** — Repository is 13 days old, v0.1.0, and private; insufficient production evidence.

14. Missing evidence needed

- **Screenshots/demo video** of the editor interface, timeline, and output previews
- **User testimonials** or case study from content creators using the tool
- **Performance metrics** (export time, frame rate, memory usage on various video lengths)
- **Accuracy metrics** for background removal and auto-reframing (comparison vs. competitors)
- **Production deployment confirmation** (current staging URL; whether emotio/sentiment inference is live)
- **Quality samples** exported from the tool showing real-world editing results
- **Architecture diagram** showing Convex backend, Inngest workflows, S3 storage, and client flow
- **Emotion/sentiment analysis validation** (precision/recall, model version, inference latency)
- **Browser compatibility matrix** (Chrome, Firefox, Safari, Edge; OS support)
- **Load testing results** (concurrent users, max video resolution/duration supported)

15. Final confidence score

Confidence: 45/100

Rationale:

1. **Strong technical foundation but unproven AI:** The project demonstrates solid engineering with React, WASM, FFmpeg, and cloud integration. However, core AI features (background removal, face detection, emotion analysis) are either algorithmic heuristics or missing implementation details. Claiming "AI-powered" would be misleading without clarification. Emotion/sentiment inference is entirely absent from the visible codebase, suggesting server-side or incomplete implementation.
2. **Early stage with no production validation:** At v0.1.0 and only 2 weeks old, there is no evidence of real user feedback, performance validation, export quality samples, or production deployment. The private repo and staging URL block public verification. Claims about accuracy, reliability, or production-readiness cannot be supported.
3. **Feature completeness gaps:** Key features like dark mode are removed, mobile support is blocked, real-time collaboration is absent, and color grading is mentioned in keywords but not implemented. Timeline

analysis and emotion/sentiment features are only partially visible, raising questions about readiness. The gap between marketing potential and shipped functionality is significant.

To improve confidence to 70+: Provide production screenshots, export samples, emotion/sentiment inference validation, performance benchmarks, user testimonials, and clarify which AI features use heuristics vs. ML models.